

Joshua Su

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EDUCATION

Georgia Institute of Technology

Bachelors of Science in Electrical Engineering, GPA: 3.85

August 2024 – Expected May 2028

Atlanta, GA

SKILLS

Engineering | RADAR, Antenna Engineering, RF Design, PCB Design, Signal Processing, Embedded Systems, FPGAs, Oscilloscopes, Logic Analyzers

Software & Programming | ANSYS Electronics Desktop, Altium Designer, Altair FEKO, Cadence Virtuoso, KiCAD, Python, C/C++, MATLAB, VHDL

EXPERIENCE

Antenna/RF Design Engineer — GT Experimental Rocketry - Ground Systems August 2026 – Present

Antenna Design | Altair FEKO

- Engineered a two-way RF ground station system for rocket-to-mission-control telemetry, designed to eliminate prior-launch communication dropouts with a modeled +75dB link margin
- Validated antenna link performance ahead of fabrication, confirming far-field radiation patterns met mission link-budget requirements through full rocket 3D mesh and test-site topology simulations in Altair FEKO
- Determined external RF hardware architecture (receive antenna, 19dB-gain LNA, bias tee injector, SX1262 LoRa transceiver (440MHz/62.5kbps), Orange Pi 5 UART-to-network bridge, WiFi point-to-point module

Hardware Engineer — GT Yellow Jacket Space Program - Avionics August 2026 - Present

PCB Design | Altium Designer

- Designed a 6-channel avionics sensor interface board in Altium Designer, implementing a 24V input protection stage (fuse, TVS diode, bulk capacitance) and a buck/LDO power tree delivering regulated 5V and 3.3V rails to downstream digital and analog circuitry
- Developed a hierarchical schematic and SPI-based signal chain connecting an STM32 microcontroller to an external ADC, enabling digitization of six 4-20mA current-loop sensor channels for pressure, temperature, and force measurement applications
- Laid out PCB with room-based floorplanning, sizing power-rail trace widths for current capacity, isolating short analog sensor traces from noisy high-current paths, and stitching ground-pour copper for signal integrity

PROJECTS

X-Band Phased Array RADAR Board July 2026 - Present

Antenna & PCB Design | ANSYS Electronics | Altium Designer

- Designed a 10.5GHz 4x4 Microstrip patch antenna phased array in ANSYS Electronics, achieving a peak realized gain of 17.3dBi at boresight, 3dB beamwidth of 13°, and a S_{11} return loss of -24dB at 10.5GHz with a 700MHz (-10dB) bandwidth
- Building RF module PCB in Altium Designer to integrate antenna array, Software Defined Radio, and microcontroller subsystems
- Achieved 60° electronic beamsteering across all 16 array elements via 4 integrated beamformer ICs, supported by an 8-layer PCB stackup isolating RF, digital, and power domains
- Designed impedance-matched, length-matched GCPW and symmetrical stripline traces for high-speed RF signal routing, minimizing crosstalk and coupling while eliminating phase delay discrepancies introduced by beamformer traces

Embedded Crash Detection Device December 2025

Embedded Systems | ESP32, C++

- Engineered a wearable/mountable crash and fall detection device using ESP32 and MPU6050 accelerometer, capable of autonomously identifying high-impact events
- Designed a modular C++ architecture separating data collection, crash-detection logic, and WiFi notification, implementing a multi-stage algorithm that analyzes pre-impact free-fall and post-impact motion stillness to compute a collision likelihood score
- Achieved 90% detection accuracy and 2% false positive rate validated through 10-foot drop tests, bicycle impact testing, and routine exercise

RELEVANT COURSEWORK

RADAR and Electromagnetic Sensing, Electromagnetics and Microwave Applications, Electromagnetics, Signals and Systems, Signal Processing, Microelectronic Circuits, Programming HW/SW Systems, Digital Design Lab